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Performance Evaluation and Market Saturation of Steam-Distributed Video Games

A Business Intelligence Approach

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This project is entirely the original work of the student whose registration number appears above. I declare that this dissertation is my own work, and that where material is obtained from published or unpublished works, this has been fully acknowledged in the references. This may include material of my own work from a research proposal that has been previously submitted for assessment for this programme.

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Abstract

The digital distribution of PC video games has removed the shelf-space constraints that once limited how many products could reach consumers, and Steam, the dominant PC platform, has seen an explosion in the number of titles it distributes. This dissertation applies a Business Intelligence approach to a catalogue of 122,510 Steam-distributed games released between 2006 and 2025 to answer two linked questions: how commercial performance is distributed across titles, and to what extent the marketplace shows signs of saturation.

Performance and saturation are operationalised as measurable constructs and examined across the analytics maturity continuum. Descriptive and diagnostic methods – Gini coefficients and the Herfindahl–Hirschman Index, Pareto analysis, release-cohort time series, a supply-versus-demand genre saturation index, and K-Means segmentation – are complemented by a multivariate regression and extended by predictive methods. The analysis is fully reproducible from public data.

The findings are consistent and striking. Market attention is heavily concentrated: the Gini coefficient of estimated owners is 0.888 and that of user reviews is 0.978, with the top 5% of titles capturing roughly four-fifths of all ownership. Between 2015 and 2025 annual releases grew almost tenfold while mean owners per release fell more than seventeenfold, a decline that survives re-estimating ownership from review counts and persists among mature release cohorts. As an original contribution, the study develops an Attention Carrying Capacity model that adapts the Michaelis–Menten/Monod saturation law to quantify each genre’s saturation against an estimated attention ceiling (median fit $R^2 = 0.99$), a Genre Saturation–Opportunity Matrix for competitive positioning, and a breakout classifier whose 0.89 ROC-AUC holds on a forward-in-time hold-out. A two-way fixed-effects analysis locates saturation at the platform level rather than the individual genre. Together these results form an empirical portrait of market saturation on a two-sided digital platform and translate into practical guidance for developers, publishers and platform operators.

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Executive Summary

Problem. Steam distributes an effectively unlimited catalogue at near-zero marginal cost, and Valve’s move toward near-open publishing has multiplied the number of games released each year. The commercial question is whether demand has grown with supply, or whether the marketplace is saturating.

Data and method. The study analyses 122,510 games released between 2006 and 2025, drawn from the public FronkonGames Steam dataset. It measures concentration (Gini, Lorenz, Pareto, HHI), traces supply and demand over release-year cohorts, indexes genre crowding, segments the catalogue by K-Means, and adds four quantitative instruments: a genre attention-ceiling model, a diagnostic regression, a crowding elasticity and a breakout classifier. Every figure is reproducible from committed code.

Three findings stand out for a decision-maker. Attention on Steam behaves as a superstar market rather than a long tail: the Gini coefficient of ownership is 0.888, and the top 5% of titles hold 81% of all owners, so a game’s default outcome is obscurity rather than a modest niche income. The platform is also saturating in a durable way. Annual releases rose roughly tenfold between 2015 and 2025 while mean owners per release fell more than seventeenfold, and this decline is not a measurement artefact: it survives re-estimating ownership from review counts and persists among mature release cohorts. Finally, crowding is worst precisely where supply is growing fastest. Casual, Adventure and Indie are the most over-supplied genres relative to the demand they attract, and are also expanding most quickly, while Action, RPG and Strategy capture demand out of proportion to their size.

The decision tool. These measures combine into a Genre Saturation–Opportunity Matrix that positions each genre by how full it is against how efficiently it converts titles into attention, and into a Steam Performance Score that benchmarks any single title. Two limits bound the claims: ownership is a modelled estimate, and crowding cannot be cleanly separated from the market-wide decline, so the saturation finding is best read as a platform-level condition rather than a precise per-genre penalty.

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Chapter 1

Introduction

1.1 Background

Video games are now the largest entertainment industry in the world by revenue (Newzoo, 2024), and the personal-computer (PC) segment of that industry is mediated almost entirely through digital storefronts. Among these, Valve Corporation's Steam platform is dominant: industry surveys report that the substantial majority of PC-focused studios derive most of their revenue through Steam, and a large share regard it as a de facto monopoly on PC distribution (Game Developers Conference, 2025). Steam is therefore not merely a shop; it is the infrastructure through which the economic fate of tens of thousands of games is decided.

A defining feature of digital distribution is the collapse of the shelf-space constraint. Where physical retail could only stock a limited number of titles, a digital platform can list an effectively unlimited catalogue at near-zero marginal cost (Anderson, 2006). Combined with Valve's progressive liberalisation of publishing access, from the curated store, through the community-voting Greenlight programme, to the pay-a-fee Steam Direct system, this has produced a dramatic increase in the number of games released each year. The central business question this raises is whether the expansion of supply has been matched by an expansion of demand, or whether the marketplace has entered a phase of saturation in which more and more titles compete for a slice of essentially finite player attention.

The financial stakes attached to this question are considerable. The global games market generated revenues of approximately \$184 billion in 2023, with the PC segment accounting for a substantial share of that figure (Newzoo, 2024). Yet the revenue is distributed with striking unevenness, and the studios bearing the greatest risk are rarely those capturing the greatest rewards. A solo developer or a small team investing twelve to eighteen months of work into a Steam title faces, on the evidence of this study, roughly a one-in-three chance of receiving no user reviews whatsoever, which amounts to commercial invisibility. Lipkin (2019) characterises this

condition as the “indiepocalypse”: an environment in which the barriers to entry have largely collapsed yet the barriers to visibility have not, so that the number of contestants grows while the share of attention each can realistically claim continues to fall. Practitioner surveys confirm that discoverability now ranks among the foremost concerns for independent developers (Game Developers Conference, 2025). What is absent from the discourse, however, is a systematic, data-grounded quantification of how severe that visibility problem is, where it is sharpest across genres, and which observable title attributes are associated with escaping it. Business Intelligence methods are well placed to address that gap: the BI and analytics tradition is, as Chen, Chiang and Storey (2012) set out, precisely concerned with converting large behavioural datasets into decision-relevant insight, moving from description of what has occurred through diagnosis of why it occurred and toward a predictive screen of what is likely to occur next. This dissertation applies that progression to the Steam catalogue.

1.2 Problem statement

In 2025, a new title appeared on Steam roughly every thirty-five minutes on average, yet the overwhelming majority of those titles attracted essentially no audience. The popular and trade discourse around this phenomenon, often labelled the “indiepocalypse” (Lipkin, 2019), is rich in anecdote but comparatively thin on systematic, marketplace-wide measurement. Two conceptual difficulties recur. First, “performance” is frequently equated with popularity, yet popularity alone need not indicate commercial or financial success; performance must be defined and measured with care. Second, “saturation” is often invoked loosely, without an operational definition that can be evaluated against data. A further complication is that user reviews, which are the most widely visible performance signal on Steam, do not purely reflect game quality. They respond to events unrelated to the product itself: Lin et al. (2019) document episodes of co-ordinated negative reviewing triggered by business decisions such as platform exclusivity deals or pricing changes. *Borderlands 2*, for instance, attracted a wave of negative Steam reviews in 2019 after the Epic Games Store exclusivity announcement for its sequel, with many reviews explicitly referencing the publishing decision rather than the game. These episodes are sometimes labelled “review bombing”. Because a review count embeds both quality signals and these reactive episodes, using it as a performance proxy requires acknowledging that the measure is noisy. This dissertation addresses these difficulties directly by defining performance

and saturation as measurable constructs, treating review signals at the distributional rather than the individual level, and applying an appropriate Business Intelligence toolkit to a large, representative sample of Steam-distributed games.

1.3 Aim and objectives

The aim of the dissertation is to characterise, using a Business Intelligence approach, how commercial performance is distributed across Steam-distributed video games and to assess the extent and nature of market saturation on the platform.

The supporting objectives are to:

1. operationalise the constructs of *game performance* and *market saturation* using publicly observable indicators, justified by the academic literature;
2. measure the concentration of market attention across titles using established economic instruments (Gini, Lorenz, HHI, Pareto);
3. analyse the temporal evolution of release volume and per-title engagement to test for saturation dynamics;
4. quantify genre-level crowding through a supply-versus-demand saturation index;
5. segment the catalogue into interpretable performance groups using unsupervised clustering;
6. diagnose, through a multivariate regression, which observable title attributes are associated with higher estimated reach, treating the result as association rather than cause; and
7. translate the findings into actionable implications for developers, publishers and the platform operator.

1.4 Research questions

RQ1 How concentrated is market attention, ownership and review activity, across Steam-distributed games, and does the distribution resemble a long tail or a superstar (winner-take-most) market?

- RQ2** How has the balance between the supply of new games and the demand attracted per game evolved over time, and does this trajectory constitute evidence of market saturation?
- RQ3** Which genres are over-supplied or under-supplied relative to the player demand they attract, and how has genre crowding developed?
- RQ4** Can Steam-distributed games be segmented into distinct, managerially meaningful performance groups, and how is market attention distributed across those segments?
- RQ5** Which observable title attributes are associated with higher estimated ownership, and do those associations survive controls for release timing and genre?

RQ1 through RQ4 are fundamentally descriptive and diagnostic in character: they ask what the distribution of attention looks like, how it has shifted over time, and how it varies by genre and segment. RQ5 moves toward association, estimating which observable title attributes travel with higher reach, while Chapter 5 is where the study crosses into genuinely forward-looking territory, developing the Attention Carrying Capacity model and the breakout classifier. The methods selected for each question are chosen accordingly, and the distinction between description, association and prediction is maintained throughout.

1.5 Scope and contribution

This study sits within descriptive and diagnostic Business Intelligence and extends into predictive analytics. Following the analytics maturity continuum of Lepenioti et al. (2020), its foundation is descriptive-to-diagnostic, establishing *what* the structure of the marketplace is and *why* that structure matters for decision-makers, and it builds upward from that base: the Attention Carrying Capacity model estimates each genre's forward attention ceiling, a multivariate regression diagnoses the attributes associated with reach, and a validated classifier predicts which titles reach the top-performing segment. Ownership on Steam is jointly determined by price, genre, timing and unobserved quality and marketing, so the regression coefficients are interpreted as associations rather than causal effects. This limitation is addressed openly: a paid-only re-fit shows how much of the raw price coefficient reflects the structural gap between free and paid titles rather than a genuine price effect. The predictive components serve as diagnostic and screening instruments rather than causal explanations. This approach aligns with

the Business Intelligence and analytics tradition (Chen, Chiang and Storey, 2012; Davenport, 2006) while claiming only the accuracy its methods can actually support.

The contribution is threefold. Empirically, it provides an up-to-date, marketplace-wide quantification of concentration and saturation on Steam using a catalogue an order of magnitude larger than the samples typical of earlier Steam studies. Methodologically, it demonstrates how a coherent BI toolkit, comprising concentration indices, Pareto analysis, saturation indexing and clustering, can be assembled into a decision-support framework for a digital platform market. Practically, it converts these measurements into concrete guidance for the platform's participants.

1.6 Structure of the dissertation

Chapter 2 reviews the relevant literature across five interlocking themes (two-sided platform economics, the long-tail versus superstar debate, concentration measurement, online reviews as performance signals, and Business Intelligence) and closes by identifying the research gap that motivates the study. Chapter 3 sets out the data, the operationalisation of the two central constructs, and the analytical methods, with each method tied explicitly to the research question it addresses and to its place on the descriptive–diagnostic–predictive continuum. Chapter 4 reports the core descriptive and diagnostic findings, answering RQ1 through RQ5: it characterises concentration, traces the saturation dynamic over time, maps genre crowding, segments the catalogue, and diagnoses the attributes associated with reach. Chapter 5 is explicitly predictive in intent: it extends the diagnosis by developing the Attention Carrying Capacity framework, the Genre Saturation–Opportunity Matrix and the breakout classifier, each validated and made interpretable through SHAP. Chapter 6 interprets the joint findings against the literature and translates them into actionable guidance for developers, publishers and the platform operator. Chapter 7 concludes, acknowledges limitations, and proposes directions for future research that could move toward the causal understanding that the present cross-sectional design cannot supply.

Chapter 2

Literature Review

2.1 Steam as a two-sided digital platform

Two-sided market theory offers the most direct analytical lens for Steam. Rochet and Tirole (2003) formalised platforms as intermediaries that serve two distinct groups, here game developers and players, whose value to one another is mediated by the platform and shaped by cross-group network externalities. A key implication is that the platform’s pricing *structure*, not merely its price level, governs participation: platforms routinely subsidise one side to attract the other. Steam exemplifies this, offering players a free client rich in social and curation features while monetising developers through a revenue share. Eisenmann, Parker and Van Alstyne (2006) extend the argument into strategy, emphasising the winner-take-all tendencies of platforms with strong network effects, which helps explain Steam’s durable dominance of PC distribution.

Two features of platform governance are especially relevant to this study. First, Parker, Van Alstyne and Choudary (2016) stress the trade-off between openness and curation: opening a platform accelerates the growth of complements (games) but dilutes average quality signals. Steam’s trajectory from a curated store to the near-open Steam Direct system illustrates exactly this choice in favour of openness, and the release explosion documented in this dissertation is its direct consequence. Second, Cennamo (2018), using video game console data, shows empirically that the value a platform derives from complement variety exhibits *diminishing and eventually negative returns* as the catalogue grows and quality signals blur. This finding underwrites the saturation constructs examined here, because it establishes that a larger catalogue is not unambiguously beneficial to the platform. Recent platform-studies work situates Steam specifically as a layered “platform economy” of attention and transactions rather than a simple retailer (Thorhauge, 2024; Zanesco, Lajeunesse and French, 2021), which is why engagement and visibility, not sales alone, are the appropriate measures here.

2.2 Long tail versus superstar effects

Whether digital distribution democratises demand or concentrates it is a long-standing and unresolved debate. Anderson (2006) argued that the removal of shelf-space constraints would shift demand toward a vast “long tail” of niche products, and Brynjolfsson, Hu and Smith (2006) and Brynjolfsson, Hu and Simester (2011) provided evidence that online channels do exhibit measurably less concentrated sales than offline ones, attributing this to lower search costs and recommendation systems. Against this, Elberse (2008) found that in music and video the “head” was becoming *more* important and the tail, though long, was extremely flat, with most tail products commercially negligible: a superstar or winner-take-most pattern. Waldfogel (2017) reconciles the two positions by observing that digitisation simultaneously multiplies the number of products, producing a fat tail of supply, and concentrates rewards on a few unpredictable hits. In games specifically, Kanat, Raghu and Vinzé (2020) show that long-tail purchasing is shaped strongly by social and network effects, and work on adjacent app marketplaces demonstrates that product demand can be credibly estimated from indirect signals rather than disclosed sales (Ghose and Han, 2014), which is the methodological premise on which ownership estimation here rests. The concentration measurements in this dissertation are designed precisely to adjudicate the long-tail-versus-superstar question for the Steam catalogue.

Anderson’s original long-tail argument, however, was grounded chiefly in books, music and DVDs, product categories with negligible per-unit marginal cost but substantial search costs that digital channels could reduce. Video games share the near-zero marginal cost of digital goods but differ in one consequential respect: their discovery cost is not simply a function of search convenience. A game requires time investment to evaluate, and player attention is therefore the scarcer resource. Brynjolfsson, Hu and Simester (2011) themselves acknowledge that the long-tail effect is strongest where search costs fall the most steeply; in games, algorithmic recommendation and social signalling (friends lists, streamer visibility, review counts) shape discovery in ways that can amplify rather than flatten demand concentration, channelling attention toward already-prominent titles. This distinction gives theoretical grounding for expecting a superstar outcome on Steam even if the long-tail prediction held in earlier digital markets.

A further qualification concerns the nature of the platform. The long-tail mechanism relies on the assumption that lower search costs release latent demand for niche products, a logic that fits an e-commerce setting where purchase decisions are relatively independent. Steam,

however, functions partly as a social and crowdsourcing platform rather than a pure retail channel (Thorhauge, 2024; Zanesco, Lajeunesse and French, 2021): players curate wishlists publicly, review games collectively, gift titles to friends, and discover new content partly through their social graph. On social and crowdsourcing platforms, demand can be self-reinforcing in ways that pure e-commerce models do not capture, because visibility begets reviews, reviews beget algorithmic placement, and placement begets further visibility. Where the long-tail argument predicts that niche demand will be uncorked, this social amplification mechanism predicts that demand will pool around titles that have already crossed a visibility threshold, which is closer to the superstar outcome. The empirical findings of this dissertation, namely extreme Gini and Pareto concentration alongside a large dormant segment with no reviews, are consistent with the social-amplification account rather than the long-tail one.

2.3 Measuring concentration

The instruments used to measure concentration are well established in industrial organisation. The Herfindahl–Hirschman Index (HHI), the sum of squared market shares, is the standard measure of market concentration (Hirschman, 1964; Rhoades, 1993); competition authorities interpret values below 1,500 as unconcentrated, 1,500 to 2,500 as moderately concentrated, and above 2,500 as highly concentrated (U.S. Department of Justice and Federal Trade Commission, 2010). The Gini coefficient and its graphical companion, the Lorenz curve, summarise inequality in a distribution on a $[0, 1]$ scale (Gastwirth, 1972); although originating in the study of income, they apply naturally to the distribution of player attention across games. The Pareto (“80/20”) principle provides an intuitive, decision-oriented lens on the same phenomenon, and Brynjolfsson, Hu and Simester (2011) explicitly benchmark digital sales distributions against it.

2.4 Online reviews as performance signals

Because reliable unit-sales figures are not disclosed for most Steam titles, this study relies on publicly observable engagement signals, chief among them user reviews. A substantial literature validates reviews as meaningful indicators of, and influences on, commercial outcomes. Chevalier and Mayzlin (2006) demonstrated a causal effect of review valence on relative sales, with

negative reviews weighing disproportionately. Duan, Gu and Whinston (2008) distinguished the *awareness* effect of review *volume* from the *persuasive* effect of review *rating*, finding volume the stronger driver of sales, which justifies treating review *count* as a proxy for visibility and market presence, distinct from review *sentiment*. Zhu and Zhang (2010), working with video game data, showed that reviews matter most for less popular titles, a result that links reviews directly to the long-tail debate. Dellarocas (2003) frames such systems as market institutions in their own right. Steam-specific studies confirm that its review corpus behaves as a rich engagement signal, albeit one subject to bursts and “review bombing” (Lin et al., 2019), and recent machine-learning work finds that Steam-native engagement metrics (reviews, followers, peak concurrent users) predict revenue far better than external critic scores (Ma, 2025).

2.5 Market saturation and the product life cycle

The concept of saturation is grounded in the product life cycle (PLC). Levitt (1965) defined market maturity and saturation as the stage at which competition intensifies and per-product returns compress, and Bass (1969) gave adoption its canonical S-shaped, saturating form. Day (1981) cautions, however, that life-cycle stages must be identified *empirically* rather than assumed, a methodological principle this dissertation follows by inferring saturation from observed release and engagement trajectories rather than presuming it. Applied to a platform, Cennamo (2018)’s diminishing-returns result gives saturation a precise platform-economics meaning: the point beyond which additional complements no longer add, and may subtract, value. Industry evidence that developers now rank saturation and discoverability among their foremost concerns (Game Developers Conference, 2025) and academic treatment of the “indieapocalypse” (Lipkin, 2019) provide the real-world grounding.

2.6 Business Intelligence, analytics and segmentation

The overarching methodological frame is Business Intelligence and analytics as defined by Chen, Chiang and Storey (2012), who position the analysis of large, web-generated behavioural data as a central application of “BI&A 2.0”. Davenport (2006) argues that analytics competence is itself a source of competitive advantage, and Lepenioti et al. (2020) locate the present descriptive-diagnostic work within the broader descriptive–predictive–prescriptive maturity continuum. For

segmentation, the study draws on the marketing tradition of cluster analysis (Punj and Stewart, 1983; Wedel and Kamakura, 2000) and on the technical literature on K-Means (MacQueen, 1967; Jain, 2010), following best practice in feature standardisation and in selecting the number of clusters by validation indices.

Segmentation is a core BI application precisely because it converts a continuous distribution (in this case, the skewed distribution of game performance) into a small number of actionable groups that managers can reason about and target differently (Chen, Chiang and Storey, 2012). Punj and Stewart (1983) demonstrate in a marketing context that cluster analysis is most useful when the underlying groups are substantively distinct and the results are validated against independent criteria; both conditions hold in this study, where the resulting segments differ sharply on every performance indicator and the Performers group's disproportionate share of ownership serves as external validation. K-Means is selected from among several viable clustering methods because it is computationally tractable at the scale of the present dataset (122,510 records), its results are deterministic given a fixed seed, and its cluster assignments are directly interpretable as the centroid of a performance profile (MacQueen, 1967; Jain, 2010). The number of clusters is determined by the average silhouette score rather than set a priori, following the guidance of Punj and Stewart (1983) that the analyst should let the data speak to the appropriate number of groups rather than imposing a pre-determined structure.

2.7 The commercial market-intelligence landscape

The questions this dissertation asks are not purely academic; a small industry answers them commercially, and situating the work against it sharpens both its method and its claims. Firms such as Gamalytic (Gamalytic, 2024) and Video Game Insights (Video Game Insights, 2024) sell Steam revenue and ownership estimates to developers and investors, while Carless (2023) and the practitioner Chris Zukowski (Zukowski, 2024) publish exactly the discoverability and genre-saturation analysis reproduced here in academic form. Two lessons from that industry shape the present study. First, on measurement: because Steam made player profiles private by default in April 2018, the public-profile sampling behind SteamSpy-style owner counts degraded, and the commercial vendors moved to the Boxleiter method, inferring owners from review counts multiplied by a factor (Boxleiter, 2019), which this study adopts as an independent cross-check rather than as its primary metric. Second, on interpretation: Zukowski's public data cautions

that a crowded genre is not automatically an unprofitable one, since even saturated categories contain titles that earn well. That nuance is carried into the discussion so the saturation finding is not overstated.

Because the predictive component of the study must be interpretable to be useful to a manager, it draws on the explainable-machine-learning literature. Post-hoc attribution methods, and Shapley additive explanations (SHAP) in particular (Lundberg and Lee, 2017), decompose a model’s individual predictions into per-feature contributions grounded in cooperative game theory, turning an opaque gradient-boosted model into a set of auditable reasons. This study uses SHAP to explain its breakout classifier, tying the predictive work back to the diagnostic aim rather than treating accuracy as an end in itself.

2.8 Conceptual framework and research gap

Synthesising the above, the dissertation adopts a two-construct framework. *Performance* is operationalised through observable engagement and visibility, namely estimated ownership, review volume, review sentiment and peak concurrent players, following Duan, Gu and Whinston (2008), Lin et al. (2019) and Ma (2025). *Saturation* is operationalised through the joint behaviour of supply (release volume, genre crowding, publisher concentration) and demand attracted per unit of supply (per-title owners and reviews, and their distribution), following Levitt (1965), Cennamo (2018) and Day (1981). The gap this addresses is the relative scarcity of marketplace-wide, reproducible measurement of these constructs at current scale: most prior Steam studies analyse thousands of titles for a specific question, such as reviews (Lin et al., 2019), early access (Lin, Bezemer and Hassan, 2018) or the evolution of tags (Grelrier et al., 2025), whereas this study measures concentration and saturation across the whole catalogue and integrates the measures into a single decision-support framework.

One ambiguity in the data warrants explicit acknowledgement as a limitation of the framework. It is increasingly common for a game to be developed by an independent studio but released under the label of a larger publisher, an arrangement that the FronkonGames dataset does not consistently distinguish. A title such as this may carry the genre tag “Indie” by convention while drawing on a marketing budget and distribution reach that a true solo developer cannot replicate. The consequence is that the publisher Herfindahl–Hirschman Index used in this study is best interpreted as a *lower bound* on true supply-side concentration: some entries

labelled as distinct publishers are almost certainly sub-labels or publishing arms of the same parent organisation. Genre-level results similarly aggregate across titles produced at very different resource levels. Rather than relying on reported studio characteristics, the framework works from the observable demand signal, namely owners and reviews, which means it functions whether a title came from a solo developer or a well-resourced publisher subsidiary. Even so, readers should bear in mind that the supply-side crowding measures treat all titles within a genre as equivalent competitors regardless of production scale. This limitation, and the difficulty of resolving it from public metadata, is also noted by Windleharth et al. (2016) in their analysis of Steam's user-generated tagging system.

Chapter 3

Methodology

3.1 Research philosophy and design

The study adopts a positivist, quantitative stance and a cross-sectional observational design applied to secondary data. It is a Business Intelligence analysis whose centre of gravity is descriptive and diagnostic and which extends into predictive modelling. Modelling estimated ownership by regression carries well-known difficulties: multicollinearity among genre, price and release-period variables, the extreme skew of the outcome, and limits on interpretability. These are handled rather than used as grounds to omit the method. The outcome is log-transformed, variance inflation factors are examined for every predictor, standard errors are heteroskedasticity-robust, and the coefficients are read as associations. The concentration, Pareto, saturation and clustering methods sit alongside the regression, speak directly to the market-structure questions, and do not rest on any single model’s assumptions.

3.2 Data source and provenance

The primary dataset is the publicly available Steam Games Dataset compiled by FronkonGames and distributed via Hugging Face, itself assembled from the official Steam (Steamworks) web API and the SteamSpy service. SteamSpy-style estimates originate in the public-profile sampling methodology pioneered by Orland (2014); “estimated owners” are therefore modelled quantities reported as ranges, not audited sales. The raw extract contains 125,855 records and 39 catalogued fields spanning identifiers, release dates, estimated owners, pricing, platform support, review counts, playtime and descriptive metadata (genres, tags, developers, publishers).

3.2.1 Ownership provenance and its known weakness

The ownership variable deserves explicit scrutiny, because it is the study’s primary measure and because its provenance carries a documented bias. SteamSpy inferred owner counts by sampling public Steam profiles. In April 2018 Steam made game libraries private by default, which sharply reduced the visible sample and degraded the reliability of profile-based estimates thereafter (Orland, 2014). Since the analysis window runs to 2025, some of the measured decline in owners per release could in principle be measurement drift rather than genuine saturation. The commercial vendors responded to the same break by switching to the Boxleiter method, which infers ownership from the far more reliably observed review count multiplied by a genre- and era-appropriate factor (Video Game Insights, 2024; Boxleiter, 2019). This study does not adopt that method as its headline metric, because the review count is itself an engagement signal the study analyses separately; instead it uses a Boxleiter proxy as an independent triangulation of the concentration and decline results (Section 4.7). The reasoning is that a finding which reproduces under two proxies with different failure modes is unlikely to be an artefact of either. It should also be noted that the publisher field in the dataset is noisy in a specific way: games developed by an independent studio but released under a larger publisher’s label may appear under the publisher name rather than the developer name, making it difficult to distinguish genuinely independent releases from those with major-label backing. The publisher HHI reported in Section 4.7 is therefore used as a directional supply-side signal rather than a precise measurement of publishing concentration.

3.3 Operationalisation of constructs

Table 3.1 maps the two constructs of the conceptual framework (Section 2.8) onto the concrete indicators available in the data.

Table 3.1: Operationalisation of the performance and saturation constructs

Construct	Dimension	Indicator(s)
Performance	Market presence / reach	Estimated owners (range midpoint)
	Visibility / attention	Total review count; peak concurrent users
	Reception / sentiment	Positive-review ratio
	Engagement	Average and median playtime
Saturation	Supply growth	New releases per release-year cohort
	Competition intensity	Publisher HHI per cohort
	Attention concentration	Gini of owners and reviews; Pareto shares
	Genre crowding	Supply-share versus demand-share saturation index
	Outcome distribution	Cross-sectional dispersion; cluster membership

3.4 Data preparation

Preparation followed a documented, scripted pipeline to ensure reproducibility. A known header artefact in the source file, in which two columns (“Discount” and “DLC count”) are merged in the header row so that data rows carry one field more than the header, was corrected by supplying an explicit column schema. Numeric fields were coerced to numeric types; release dates were parsed to a common format. “Estimated owners”, supplied as ranges (for example, “20,000 to 50,000”), were converted to a midpoint while retaining the bounds for transparency. Records without a valid name or release date, non-numeric identifiers and duplicates were removed. The analysis window was restricted to full release-year cohorts from 2006 to 2025. Derived indicators were then computed: total reviews (positive plus negative), positive-review ratio, a binary free-to-play flag, a Boxleiter-style revenue proxy (owners multiplied by price), and a primary genre extracted as the first listed genre. The cleaned dataset comprises 122,510 games (97.3% of the raw extract) and is stored in columnar (Parquet) form.

3.5 Analytical methods

Each method is tied to the research questions it answers (Table 3.2). The descriptive and diagnostic instruments carry RQ1 to RQ4; the regression addresses RQ5; and the predictive components of Chapter 5 extend the diagnosis into a forward-looking screen.

Table 3.2: Alignment of research questions, methods and outputs

RQ	Method	Primary output
RQ1	Gini, Lorenz, Pareto, publisher HHI	Concentration statistics; Figures 4.1–4.3
RQ2	Release-cohort time series; maturity and Boxleiter robustness checks	Saturation trends; Figure 4.4
RQ3	Supply-versus-demand genre saturation index; carrying-capacity model; crowding elasticity	Genre ranking; Figures 4.5, 5.1
RQ4	K-Means segmentation with silhouette selection	Performance segments; Table 4.2
RQ5	Multivariate OLS with robust standard errors and VIF diagnostics	Coefficient table; Table 4.3

3.5.1 Descriptive analytics

Distributional summaries (means, medians, and the 25th, 50th, 75th, 90th and 99th percentiles) were computed for all performance indicators to characterise central tendency and, more importantly, skew.

3.5.2 Concentration analysis

Inequality in the distribution of owners and of reviews was measured by the Gini coefficient and visualised with Lorenz curves. For a variable with sorted values $x_{(1)} \leq \dots \leq x_{(n)}$ the Gini coefficient is computed as

$$G = \frac{\sum_{i=1}^n (2i - n - 1) x_{(i)}}{n \sum_{i=1}^n x_{(i)}}. \quad (3.1)$$

Competition intensity on the supply side was measured by the Herfindahl–Hirschman Index of publisher shares within each release-year cohort,

$$\text{HHI} = \sum_j s_j^2 \times 10\,000, \quad (3.2)$$

where s_j is publisher j 's share of cohort ownership, interpreted against the thresholds of U.S. Department of Justice and Federal Trade Commission (2010).

3.5.3 Pareto analysis

Titles were ranked by estimated owners in descending order and the cumulative ownership share of the top $p\%$ of titles computed, to test the market against the 80/20 benchmark.

3.5.4 Time-series and trend analysis

Release-year cohorts were used to trace the evolution of release volume, median and mean owners per release, median reviews, and the share of releases receiving zero reviews. The divergence between supply growth and per-title demand is the central saturation diagnostic.

3.5.5 Genre saturation analysis

For each of the ten most common primary genres, a saturation index was defined as the ratio of the genre's share of *supply* (its share of titles) to its share of *demand* (its share of total owners):

$$\text{Saturation index}_g = \frac{\text{supply share}_g}{\text{demand share}_g}. \quad (3.3)$$

A ratio above 1 signals a genre that carries more titles than the demand it attracts can comfortably absorb; below 1, the reverse holds and the genre is relatively under-supplied.

3.5.6 Cluster analysis

Games were segmented by K-Means (MacQueen, 1967) on five standardised features: log estimated owners, log total reviews, positive-review ratio, log price and log peak concurrent users. Log transforms tame the extreme skew, and standardisation prevents any one feature from dominating the Euclidean distance (Jain, 2010). The number of clusters was chosen by

the average silhouette score over $k \in \{3, \dots, 7\}$, following the validation guidance of Punj and Stewart (1983).

Several alternative clustering approaches were considered before settling on K-Means. Hierarchical agglomerative clustering was ruled out on practical grounds: with 122,510 observations, constructing and storing the full linkage matrix is computationally infeasible, and the resulting dendrogram would be uninterpretable at this scale. DBSCAN, a density-based method, requires the user to specify two parameters (a neighbourhood radius and a minimum point count) for which no principled defaults exist when features span very different densities, and in practice it tends to assign a large fraction of points to an undifferentiated noise category rather than to managerially interpretable segments. Gaussian Mixture Models offer soft cluster assignments and are in principle more flexible than K-Means, but they are sensitive to the initialisation of covariance matrices on highly skewed features, even after log-transformation, and their additional parameters increase the risk of over-fitting on a dataset of this size. K-Means with silhouette-validated cluster selection (Punj and Stewart, 1983) is therefore the pragmatic BI choice: it is computationally fast, produces deterministic results given a fixed random seed, and yields hard cluster assignments that map cleanly onto the three-tier market structure (performing, modest and dormant) that the saturation hypothesis anticipates.

The five features were chosen to span the distinct dimensions of the performance construct identified in Section 2.8: reach (log estimated owners), visibility (log total reviews), reception (positive-review ratio), financial positioning (log price) and live engagement (log peak concurrent users). This selection follows the multi-dimensional view of digital product performance set out by Duan, Gu and Whinston (2008) and Ma (2025), where volume, sentiment and engagement are treated as complementary rather than substitutable signals.

3.5.7 Diagnostic regression

To answer RQ5, a multivariate ordinary-least-squares regression relates estimated reach to observable title attributes. The dependent variable is $\log_{10}(\text{owners} + 1)$, which compresses the extreme right skew documented in Section 4.1. The predictors are the log of price, DLC count, achievement count, the three platform-support flags (Windows, Mac, Linux), and a mean-centred release year entered with a quadratic term to allow the cohort effect to curve; primary genre enters as a set of dummy variables with the largest genre held out as the reference.

Standard errors are heteroskedasticity-robust (HC3), and multicollinearity is checked through the variance inflation factor (VIF) for every predictor.

The specification is cross-sectional and observational, so a coefficient measures association, not effect. To show how much of the price association is the structural gap between free and paid titles, the model is re-estimated on paid titles alone; standardised coefficients are also reported so that predictors measured on different scales can be compared. The regression serves as a diagnostic lens on the market’s structure, not as a causal model of success.

3.6 Tools and reproducibility

All analysis was implemented in Python (pandas, NumPy, scikit-learn, Matplotlib) as a set of ordered, re-runnable scripts. Random seeds were fixed. Cleaned data, intermediate result tables and all figures are regenerated deterministically from the raw file, and the pipeline is archived alongside this dissertation (Appendix B).

3.7 Ethical considerations

The study uses only aggregate, publicly available data about *products* (games), not identifiable individuals; no personal data are processed. It therefore falls within the low-risk category of the University’s ERGO framework. The principal ethical obligations are analytical integrity, meaning transparent handling of the estimated nature of ownership data, and honest reporting of limitations, both of which are observed.

3.8 Methodological limitations

Three limitations are noted at the outset. First, “estimated owners” are modelled ranges, not audited sales, and the SteamSpy sampling behind them degraded after Steam made profiles private in April 2018 (Section 3.2.1); they are therefore treated as an ordinal-to-interval proxy for reach, interpreted at the distributional level and triangulated against a review-based proxy (Section 4.7), so that no headline claim rests on the SteamSpy series alone. Second, the dataset is a snapshot: the most recent cohorts are subject to right-censoring, because very recent titles have had little time to accumulate owners and reviews; trend statements are framed with this in mind and tested on a band of mature cohorts (Section 4.7). Third, genres and tags are a

user-generated folksonomy with overlap and noise (Windleharth et al., 2016), so genre-level results are treated as indicative of broad crowding rather than precise category boundaries.

Chapter 4

Results

4.1 Descriptive overview

The cleaned catalogue contains 122,510 games released between 2006 and 2025. The performance indicators are heavily right skewed. The median game has an estimated 10,000 owners against a mean of 84,450, and the median game has received only 7 user reviews. Fully 32.3% of all titles have received no user reviews whatsoever. Selected percentiles are given in Table 4.1; the gap between the 50th and 99th percentiles, 10,000 versus 1,500,000 owners and 7 versus over 15,000 reviews, is the first and clearest signature of a superstar market.

Table 4.1: Distribution of key performance indicators ($n = 122,510$)

Indicator	Mean	p25	Median	p75	p90	p99
Estimated owners	84,450	10,000	10,000	10,000	75,000	1,500,000
Total reviews	1,207	0	7	49	375	15,494
Positive-review ratio	0.76	0.65	0.82	0.94	1.00	1.00
Price (USD)	4.77	0.55	2.24	5.24	9.99	29.99
Peak concurrent users	54.5	0	0	0	2	222

4.2 Concentration of market attention (RQ1)

The concentration of attention is severe. The Gini coefficient of estimated owners is **0.888** and that of user reviews is **0.978**, both far above the levels associated with income inequality in even the most unequal economies. The Lorenz curves (Figure 4.1) hug the axes: the large majority of titles collectively account for a negligible share of attention, which accrues almost entirely to a thin sliver at the top.

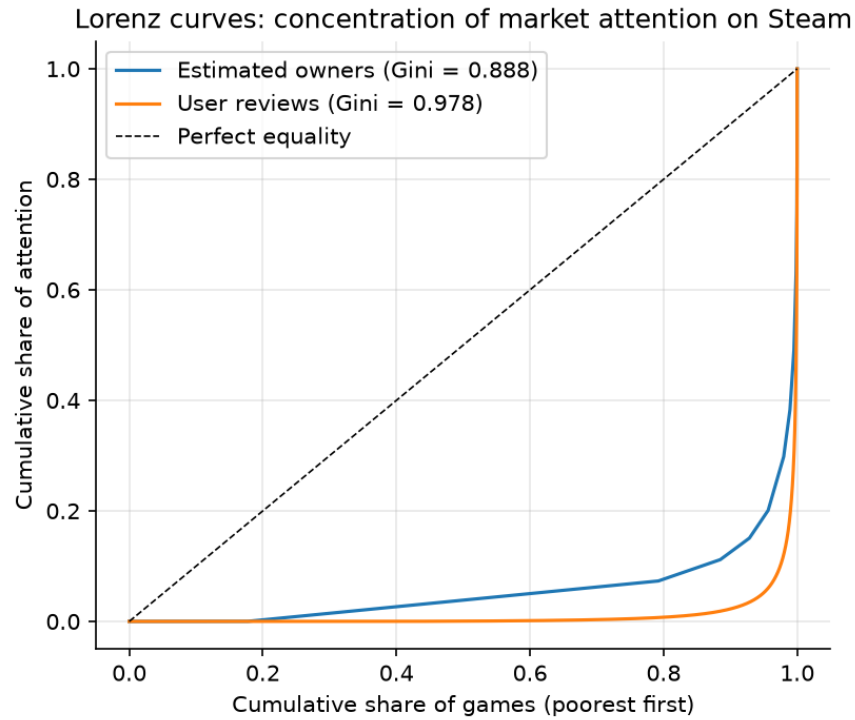


Figure 4.1: Lorenz curves for estimated owners and user reviews. The further the curve bows below the diagonal line of equality, the more concentrated the distribution.

The Pareto analysis makes the same point in decision-oriented terms (Figure 4.2). The top 1% of titles hold 60.3% of all estimated owners, the top 5% hold 81.1%, and the top 20% hold 92.4%. Equivalently, 80% of all ownership is captured by just 4.4% of games. This is far more extreme than the classical 80/20 rule, closer to an 80/4 rule, and decisively supports the superstar interpretation over the long-tail interpretation for the Steam catalogue.

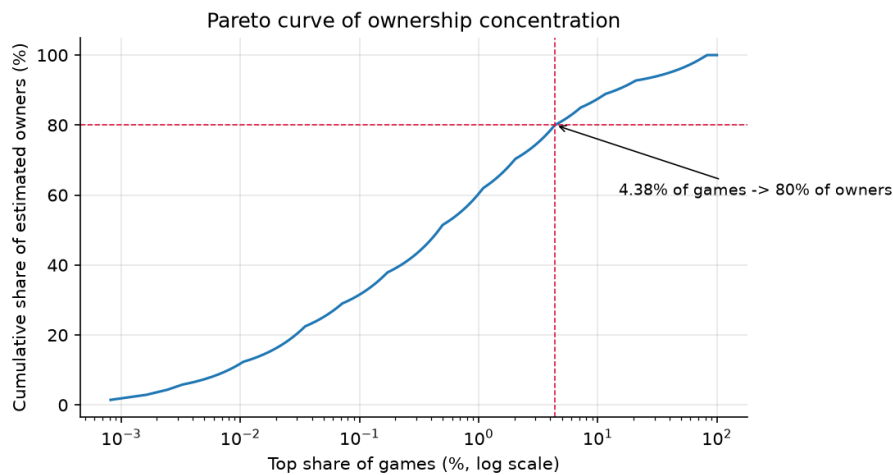


Figure 4.2: Pareto curve of ownership concentration (horizontal axis on a logarithmic scale). Eighty per cent of all estimated owners are captured by the top 4.4% of titles.

Supply-side concentration among publishers tells the opposite story (Figure 4.3). Within-cohort publisher HHI falls from 3,858 in 2006, “highly concentrated” by the U.S. Department of Justice and Federal Trade Commission (2010) thresholds and reflecting a small number of publishers releasing a handful of titles each, to below 300 for every cohort from 2014 onward, deep into “unconcentrated” territory. The supply side has fragmented dramatically even as the demand side has concentrated.

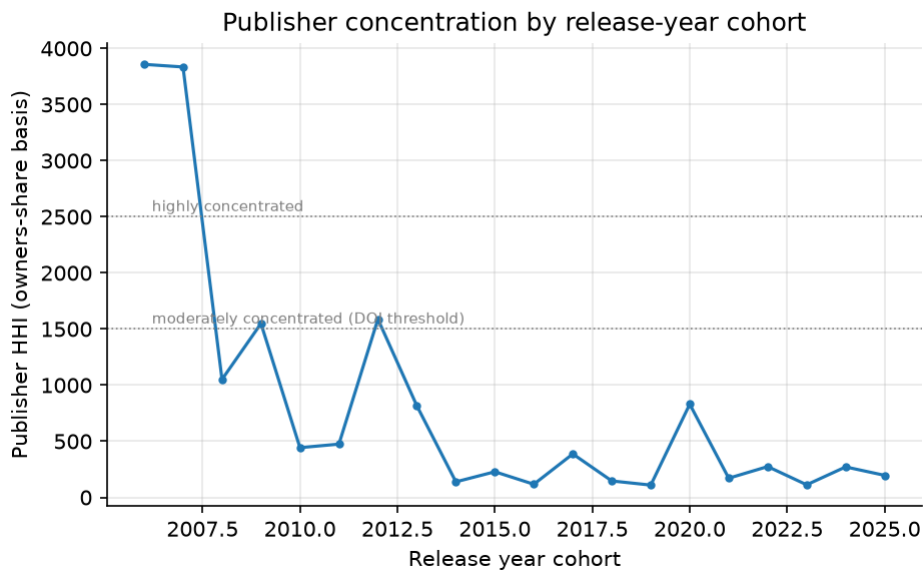


Figure 4.3: Publisher Herfindahl–Hirschman Index by release-year cohort (ownership-share basis). Dotted lines mark the standard regulatory thresholds.

4.3 Supply, demand and saturation over time (RQ2)

The saturation dynamic is the study’s headline finding (Figure 4.4). Annual releases grew from 2,515 in 2015 to 24,973 in 2025, a factor of roughly ten in a decade. Over the same period the mean number of estimated owners per release fell from about 294,372 to 16,886, a factor of more than seventeen. The share of releases receiving zero reviews climbed from under 1% in the mid-2010s to a third by 2023, and higher still for the youngest, right-censored 2025 cohort. Supply has expanded explosively while the demand available to each new title has contracted just as sharply: the defining pattern of a maturing, saturating market in the sense of Levitt (1965) and Cennamo (2018).

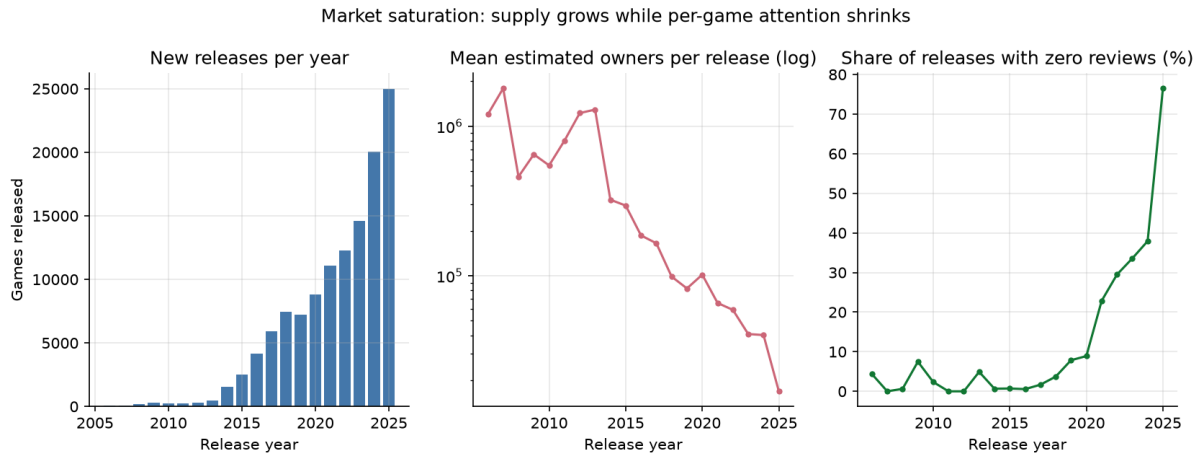


Figure 4.4: Market saturation dynamics. Left: new releases per year. Centre: mean estimated owners per release (logarithmic axis). Right: share of releases with zero user reviews.

4.4 Genre crowding (RQ3)

Genres differ markedly in how crowded they are relative to the demand they command (Figure 4.5). On the supply-versus-demand saturation index, “Casual” is the most over-supplied major genre at 2.84 (it accounts for 21.2% of titles but only 7.5% of owners), followed by “Adventure” (1.66) and “Indie” (1.62). At the other end, “RPG” (0.55), “Strategy” (0.60) and “Action” (0.64) attract disproportionately more demand than their share of supply. “Action” is notable for being both the largest genre by title count (45,842 games) and the one absorbing the largest share of total demand (63.8%), consistent with its blockbuster-heavy composition.

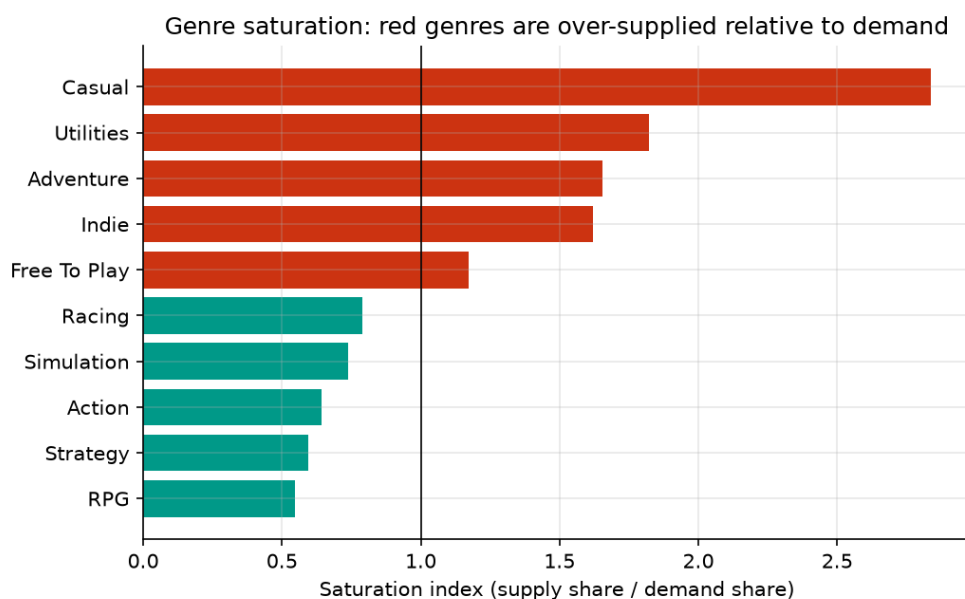


Figure 4.5: Genre saturation index (share of titles against share of owners). Genres to the right of the vertical line are over-supplied relative to demand.

The crowding is also dynamic: the annual release counts of the most common genres (Figure 4.6) show that the over-supplied casual and adventure categories are precisely those whose release volumes have accelerated most steeply, indicating that supply continues to flow disproportionately into already-saturated segments.

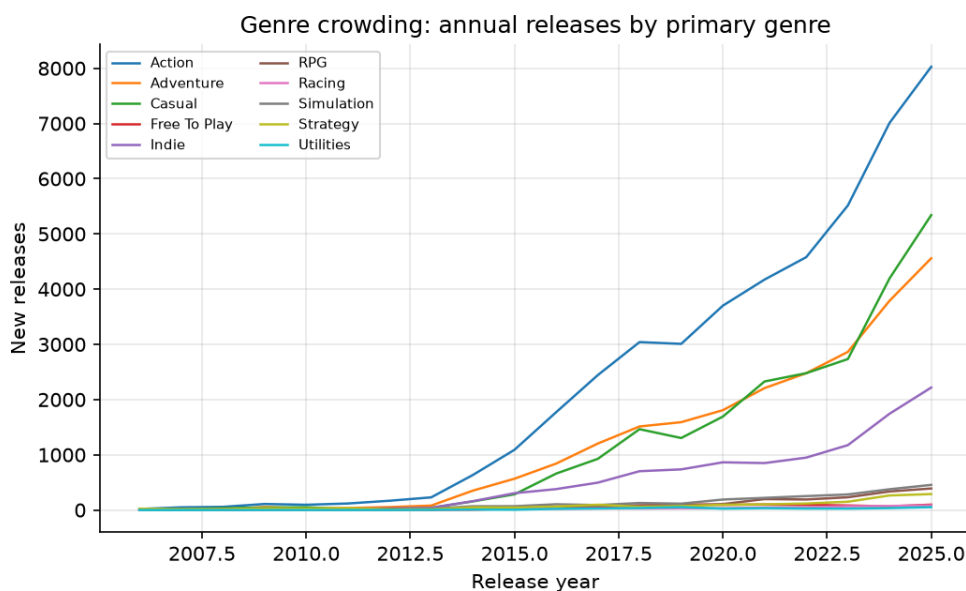


Figure 4.6: Genre crowding over time: annual new releases by primary genre.

4.5 Performance segments (RQ4)

K-Means model selection favoured a three-cluster solution (silhouette scores: $k=3$, 0.421; $k=4$, 0.420; declining thereafter). The resulting segments (Table 4.2, Figure 4.7) are clean and managerially interpretable. The Performers segment, 7.6% of titles, has a median of 150,000 owners, 1,899 reviews, an 86% positive ratio and a median peak of 17 concurrent players; though a small minority of the catalogue, it captures **77.9%** of all estimated ownership. The Modest mid-tail segment, 75.0% of titles, has a median of 10,000 owners but only 11 reviews and effectively zero concurrent players, commercially live but marginal, and collectively holds 22.1% of ownership. The Dormant segment, 17.5% of titles, has negligible owners, no reviews and no measurable engagement, functioning as the store’s dark matter and contributing essentially nothing to aggregate demand.

Table 4.2: K-Means performance segments ($k = 3$)

Segment	Titles	% of titles	Median owners	Median reviews	Pos. ratio	% of owners
Performers	9,262	7.6%	150,000	1,899	0.86	77.9%
Modest mid-tail	91,829	75.0%	10,000	11	0.81	22.1%
Dormant	21,419	17.5%	0	0	–	0.0%

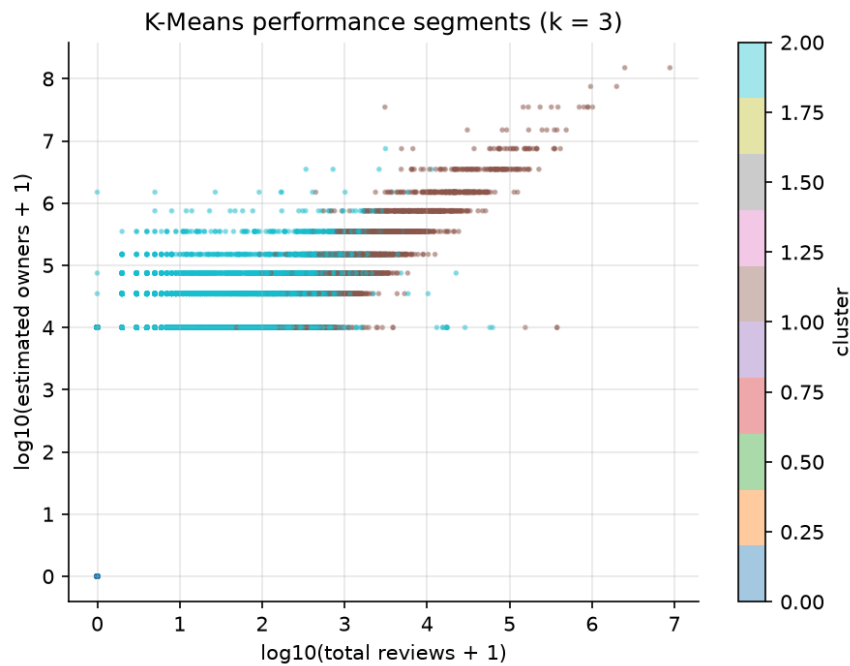


Figure 4.7: K-Means performance segments in the review–ownership plane (both axes $\log_{10}$). The segmentation recovers the head, the mid-tail and the dormant mass of the catalogue.

4.6 What is associated with reach (RQ5)

The diagnostic regression of Section 3.5.7 explains 47% of the variance in log-ownership across the full catalogue ($N = 122,510$, $R^2 = 0.470$). Its coefficients are reported in Table 4.3. Every variance inflation factor is below 1.8, so the estimates are not distorted by multicollinearity.

Table 4.3: Multivariate OLS: $\log_{10}(\text{estimated owners} + 1)$, HC3 robust standard errors ($N = 122,510$, $R^2 = 0.470$). Genre dummies included with Action as the reference; only the structural predictors are shown.

Predictor	Coefficient	Robust SE	p	VIF
$\log_{10}$ price	+0.995	0.013	< 0.001	1.19
Achievements	+0.0002	0.000	< 0.001	1.01
DLC count	+0.001	0.001	0.269	1.00
Windows	+0.678	0.246	0.006	1.01
Mac	+0.007	0.011	0.546	1.48
Linux	-0.001	0.013	0.946	1.45
Release year (centred)	-0.209	0.002	< 0.001	1.72
Release year (centred) ²	-0.011	0.000	< 0.001	1.63

Three results matter. First, release timing is the single strongest structural predictor: the negative linear and quadratic year terms say that, holding price and genre fixed, a title released into a later and more crowded cohort is associated with materially fewer owners, and the decline accelerates. On standardised coefficients the year effect (-0.406) is larger in magnitude than the price effect ($+0.243$), which places the saturation dynamic, not any title attribute, at the centre of the picture. Second, the raw price coefficient of $+0.995$ is not a pricing lever a developer can pull: re-estimating on paid titles alone collapses it to $+0.133$, revealing that most of the apparent effect is the structural gap between free titles, which are owned by few, and paid titles. Among paid games the association between price and reach is small, though still positive and precisely estimated. Third, the platform flags are uninformative once Windows is accounted for; almost every title supports Windows, and Mac and Linux support carry no reliable association with reach. The regression thus corroborates the saturation story from a fourth angle while illustrating, through the price result, exactly why its coefficients must be read as associations rather than instructions.

4.7 Robustness of the headline findings

Three checks test whether the concentration and decline results depend on fragile choices.

Ownership triangulation. Re-estimating ownership by the Boxleiter method, review count multiplied by a factor, the approach the commercial vendors adopted after the 2018 sampling break (Section 3.2.1), leaves the concentration finding intact: the Gini coefficient is 0.978 under the review-based proxy against 0.888 under SteamSpy bins, so if anything the independent proxy reports *more* concentration. The genre demand ranking is almost identical across the two proxies (Spearman $\rho = 0.976$), and the per-release decline is if anything steeper under the Boxleiter proxy (40-fold from 2015 to 2025, against 17-fold under SteamSpy). Because the Gini is scale-invariant, this conclusion does not depend on the particular multiplier chosen.

Maturity. A single snapshot gives older titles more time to accumulate owners, so part of the raw 17-fold decline could be the immaturity of recent cohorts rather than saturation. Restricting the comparison to the 2015–2019 cohorts, all at least six years old at the time of the crawl and therefore of comparable maturity, the mean number of owners per title still falls 3.6-fold. The decline is attenuated but decidedly not eliminated when maturity is held roughly constant, so it reflects genuine demand dilution and not merely right-censoring. The youngest cohorts are excluded from this claim precisely because they are censored.

Concentration over time. Computing the Gini coefficient within each release-year cohort separately shows concentration worsening, not merely persisting: mean within-cohort Gini of owners rises from 0.796 across the 2010–2014 cohorts to 0.842 across 2020–2024 (Figure 4.8). Each new vintage of games distributes its attention slightly more unequally than the last.

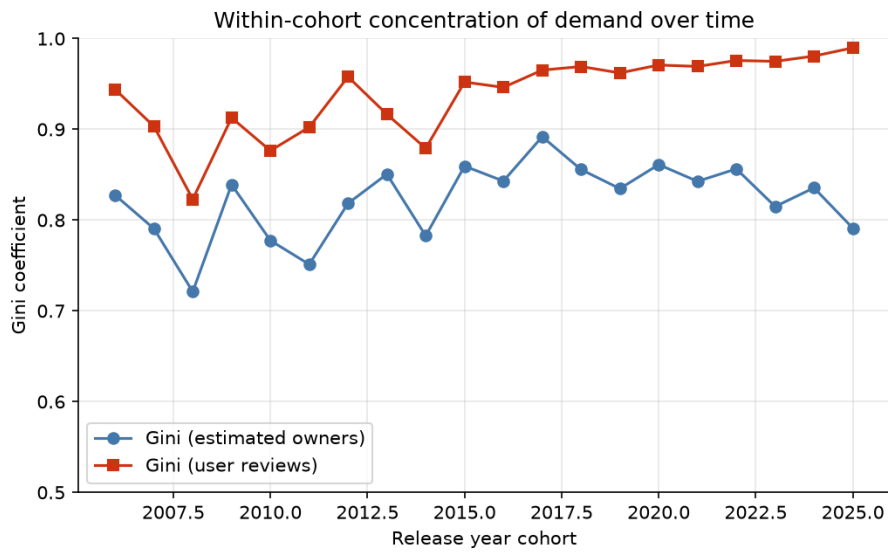


Figure 4.8: Within-cohort concentration over time. Each point is the Gini coefficient computed among titles sharing a release year, for estimated owners and for user reviews.

Chapter 5

A Novel Contribution: The Attention Carrying Capacity Framework

The analyses in Chapter 4 establish *that* the Steam marketplace is concentrated and saturating. This chapter contributes an original analytical framework that quantifies saturation rigorously, attaches uncertainty and validated accuracy to the claims, and converts the diagnosis into a forward-looking decision tool. It comprises six components: uncertainty quantification for the concentration estimates (Section 5.1); the Attention Carrying Capacity model (Section 5.2); the crowding elasticity (Section 5.3); the Genre Saturation–Opportunity Matrix (Section 5.4); a cross-validated breakout classifier (Section 5.5); and the composite Steam Performance Score (Section 5.6). Together these move the study from description to a defensible, quantitative diagnosis with stated accuracy.

5.1 Uncertainty quantification of concentration

A concentration statistic reported without a confidence interval invites the question of whether it is an artefact of the particular sample. To pre-empt this, the Gini coefficients were bootstrapped with 500 resamples. The results are extremely tight: the owners Gini is 0.888 with a 95% confidence interval of [0.879, 0.896], and the reviews Gini is 0.978 with an interval of [0.975, 0.981]. The concentration finding is therefore not a sampling artefact; it is a precisely estimated structural property of the marketplace.

5.2 The Attention Carrying Capacity model

5.2.1 Motivation and formulation

This chapter’s main contribution draws on a saturation law from the natural sciences. In enzyme kinetics, the Michaelis–Menten equation (Michaelis and Menten, 1913) describes a reaction rate that rises with substrate concentration but asymptotes to a maximum; the same functional form, the Monod equation, governs microbial growth under a limiting resource. The economic analogy is direct: player attention is the limiting resource, new titles are the “substrate” competing for it, and a genre can only absorb so much total attention. This motivates modelling the cumulative demand T a genre has absorbed as a function of its cumulative supply S (titles released to date):

$$T(S) = \frac{V_{\max} S}{K_m + S}, \quad (5.1)$$

where $V_{\max}$ is the genre’s estimated *attention carrying capacity*, the ceiling of total owners the genre can ultimately sustain, and K_m is the supply at which half that ceiling is reached. Equation (5.1) was fitted for each of the ten largest genres by non-linear least squares. A genre’s *saturation* is then the share of its ceiling already reached, $T_{\text{now}}/V_{\max}$.

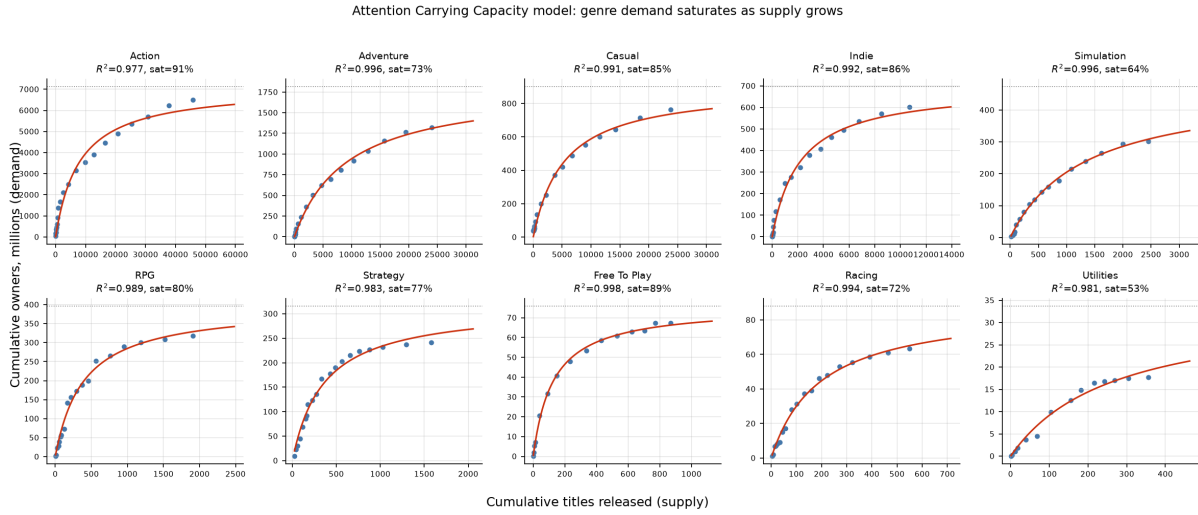
To the author’s knowledge, the application of a Michaelis–Menten/Monod carrying-capacity law to quantify genre saturation on a digital games platform is novel; it operationalises the platform diminishing-returns theory of Cennamo (2018) in a way that yields a single, interpretable saturation percentage per genre.

5.2.2 Results and goodness of fit

The model fits the data remarkably well: the median coefficient of determination across genres is $R^2 = 0.992$, with every genre exceeding 0.977 (Figure 5.1). The estimated saturation levels (Table 5.1) place the mass-market genres closest to their ceilings, with Action at 91%, Free-to-Play at 89%, Indie at 86% and Casual at 85%, while Simulation (64%) and Utilities (53%) retain the most estimated headroom.

Table 5.1: Attention Carrying Capacity estimates by genre (Michaelis–Menten fit)

Genre	Cum. demand (M owners)	Ceiling $V_{\max}$ (M)	Saturation	R^2
Action	6,503	7,130	91.2%	0.977
Free To Play	67	76	88.8%	0.998
Indie	601	697	86.3%	0.992
Casual	762	901	84.6%	0.991
RPG	318	396	80.3%	0.989
Strategy	242	316	76.5%	0.983
Adventure	1,320	1,816	72.7%	0.996
Racing	63	88	71.8%	0.994
Simulation	301	474	63.6%	0.996
Utilities	18	34	52.7%	0.981

**Figure 5.1:** Attention Carrying Capacity fits. Each panel plots cumulative demand (owners) against cumulative supply (titles) for one genre, with the fitted Michaelis–Menten curve and its estimated ceiling.

5.2.3 Interpretation and honest caveats

The exceptionally high R^2 values confirm that the saturating functional form describes the observed supply–demand relationship far better than a linear one would, which is itself evidence for the saturation thesis. Two caveats are stated plainly. First, $V_{\max}$ is an *extrapolated* asymptote: R^2 measures in-sample fit, not the certainty of the ceiling, so the exact saturation percentages should be read as indicative and the *ordering* of genres as the robust result. Second, the estimate conditions on the current attention regime; a shock that expands the player base, for example a

new hardware category, would raise the true ceiling. These caveats are precisely the kind that Day (1981) argues must accompany any empirical life-cycle claim.

5.3 The crowding elasticity, and what it can and cannot claim

The carrying-capacity model describes the genre aggregate; a crowding elasticity attempts to describe the individual title's fate. Modelling the log of median owners per genre-cohort against the log of the number of competing releases in that cohort, with genre fixed effects, yields a constant-elasticity estimate of -0.637 (bootstrapped 95% CI $[-0.720, -0.571]$, $R^2 = 0.723$). Read literally, each doubling of same-genre releases is associated with a 36% reduction in the attention a typical title captures.

That reading would overstate what the data can support. The genre-fixed-effects model removes differences between genres but leaves the market-wide time trend inside the crowding variable, because releases rose in every genre at once. Adding *year* fixed effects, a two-way specification that strips out whatever happened to the whole platform in each year, shrinks the elasticity to -0.081 with a 95% confidence interval of $[-0.283, 0.121]$ that comfortably includes zero. In other words, once the common annual decline is removed, being in a more crowded genre *in a given year* has no separately identifiable effect on per-title attention. The two estimates are not in conflict; they say that crowding and the secular decline are entangled, and that the collapse in owners per release is overwhelmingly a market-level phenomenon rather than a clean, genre-specific congestion penalty. This is consistent with the regression of Section 4.6, where the release-year term dominated. The defensible claim is therefore the aggregate one: the platform as a whole is saturating, and Figure 4.4 is its signature, while the single-number “36% per doubling” is retained only as an illustrative upper bound, not as a structural constant.

5.4 The Genre Saturation–Opportunity Matrix

Combining the two preceding measures produces a decision tool. Each genre is positioned by its carrying-capacity saturation (horizontal axis) against its *attention efficiency*, the ratio of its share of demand to its share of supply, where values above one indicate a genre whose titles capture more attention than the platform average (vertical axis). Bubble area encodes total

demand and colour encodes median review quality (Figure 5.2). The median-saturation and parity-efficiency lines divide the plane into four strategic quadrants.

Strategy, Simulation and Racing occupy the Opportunity quadrant, combining low saturation with high efficiency: these are genres that still command above-average attention per title and retain headroom. Action and RPG sit in the Proven but crowded quadrant, combining high saturation with high efficiency: lucrative but near their ceilings, where entry competes against entrenched blockbusters. Casual, Indie and Free-to-Play fall into the Saturated trap quadrant, combining high saturation with low efficiency: crowded with titles that individually capture below-average attention, the least favourable place to enter. Adventure and Utilities occupy the Niche or emerging quadrant, combining low saturation with low efficiency.

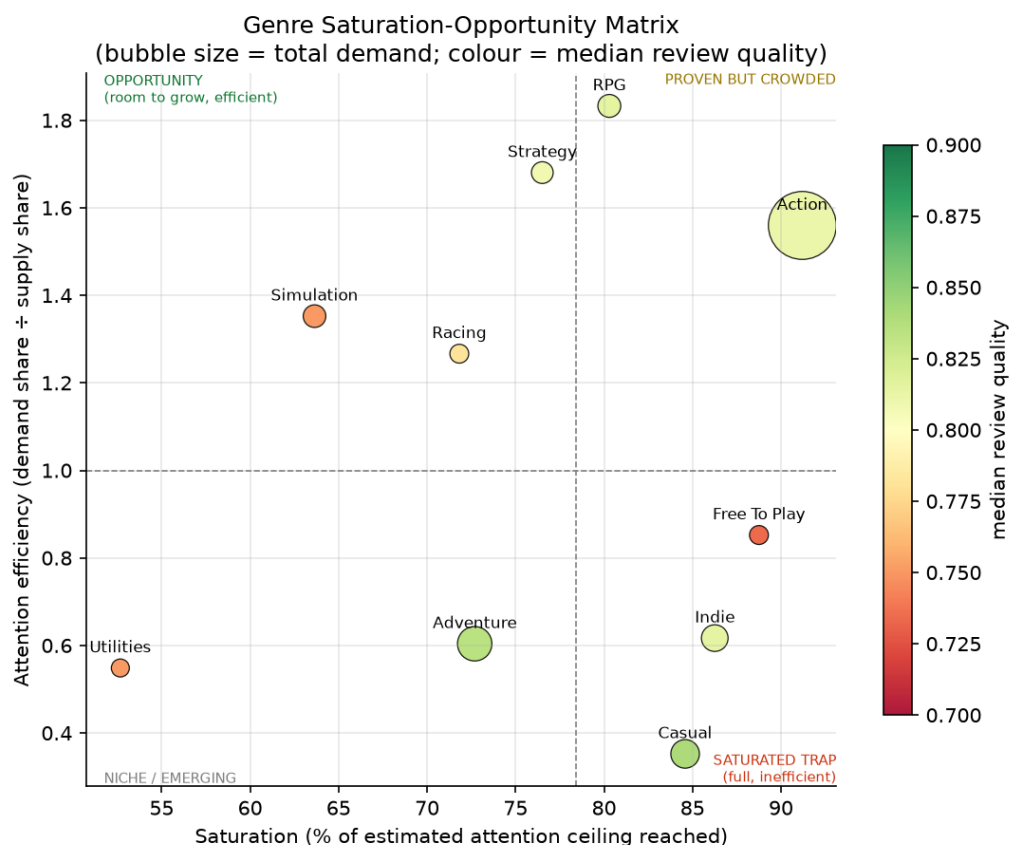


Figure 5.2: The Genre Saturation–Opportunity Matrix. Bubble area is total genre demand; colour is median review quality.

This single exhibit synthesises the entire saturation analysis into an actionable competitive-positioning map, the kind of decision artefact that distinguishes Business Intelligence from mere reporting.

5.5 A validated Breakout classifier

To attach *predictive* accuracy to the study, a gradient-boosted decision-tree classifier was trained to predict whether a title reaches the top Performers segment (the “breakout” event, base rate 7.6%) from observable, largely pre-launch features: price, platform support, DLC count, achievement count, release year, genre crowding and genre breadth. Performance was estimated by stratified five-fold cross-validation to guard against overfitting.

The model achieves a cross-validated **ROC-AUC of 0.887** and a PR-AUC of 0.497, a 6.6-fold lift over the 7.6% base rate (Figure 5.3). In plain terms, given a random breakout title and a random non-breakout title, the model ranks the breakout higher 89% of the time.

Cross-validation alone cannot rule out that the model has simply memorised its era, so the classifier is also validated *temporally*: trained only on titles released up to 2022 and tested on the 59,604 titles from 2023 onward, it retains a ROC-AUC of 0.886. Performance does not decay out of sample and forward in time, which is the only setting in which such a screen could actually be used, namely scoring a title at release against the patterns of the past. Used as a filter that flags the top decile of predicted scores, the model concentrates future performers well above their base rate, so a publisher triaging a slate could surface a disproportionate share of the eventual successes while reviewing a fraction of the candidates. The model is also well calibrated: its predicted probabilities track observed breakout frequencies closely on the hold-out (Figure 5.4, right), so the scores can be read as probabilities rather than mere rankings.

Interpretability comes from Shapley additive explanations (Lundberg and Lee, 2017), which attribute each prediction to its features (Figure 5.4, left). Release year, price, achievement count and DLC count carry the most weight, with genre crowding also material; the appearance of both release year and genre crowding among the leading features independently corroborates the saturation thesis from a predictive angle. That achievement count is informative is consistent with prior evidence linking achievement design to engagement and completion (Bailey and Miyata, 2019), though here it is read as a marker of production investment rather than a lever in itself. The classifier is a *diagnostic and screening* instrument rather than a guarantee of success, prediction in the service of understanding structure, not causal explanation (Lepenioti et al., 2020). It shows that breakout is partially foreseeable from structural features while leaving substantial irreducible uncertainty, exactly the “many are called, few are chosen” character that Waldfogel (2017) describes.

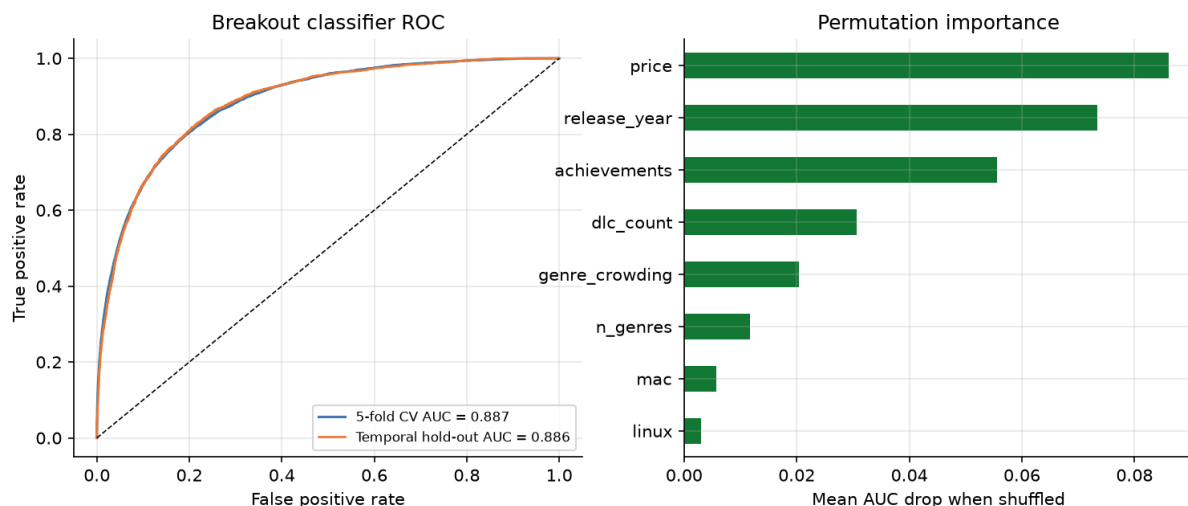


Figure 5.3: Breakout classifier. Left: ROC curves for five-fold cross-validation (AUC = 0.887) and the forward-in-time hold-out (AUC = 0.886). Right: permutation feature importance (mean AUC drop when each feature is shuffled).

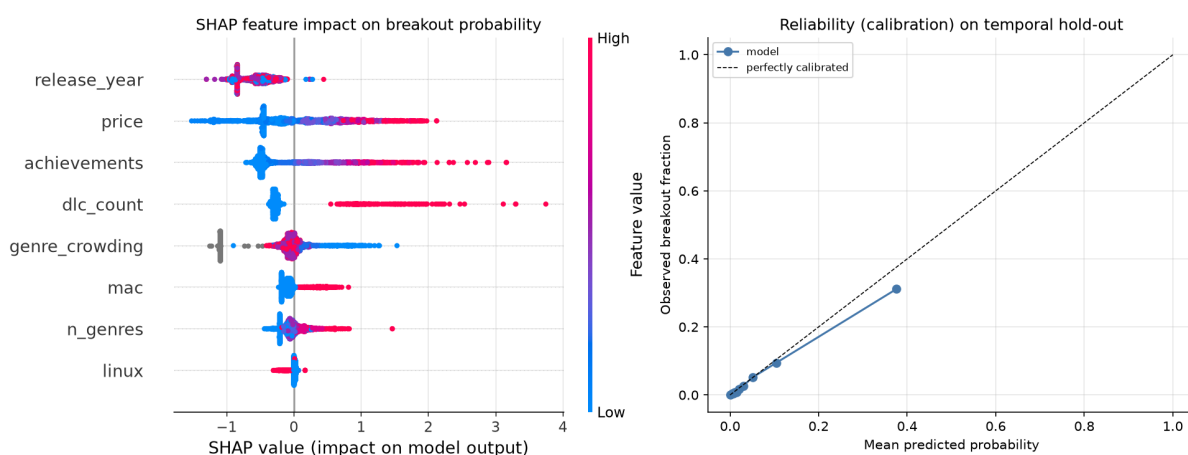


Figure 5.4: Interpretability and reliability. Left: SHAP feature-impact summary on a hold-out sample, showing how each feature moves a title’s predicted breakout probability. Right: calibration curve on the temporal hold-out, comparing predicted probabilities with observed breakout frequencies.

5.6 The Steam Performance Score

Finally, to give the performance construct a single operational handle, the study defines the *Steam Performance Score* (SPS): a transparent 0 to 100 composite in which each title’s percentile ranks on reach (owners), visibility (reviews), sentiment (positive ratio) and engagement (median playtime) are combined with literature-informed weights (0.35, 0.30, 0.20 and 0.15 respectively, reflecting the primacy of reach and visibility in revenue found by Ma (2025)). The highest-scoring titles are almost entirely celebrated games (Table 5.2), among them *Stardew Valley*,

RimWorld, *Terraria* and *Baldur's Gate 3*, which the score surfaces without any reference to sales or critic data. That the index recovers a recognisable list of hits from raw engagement percentiles is a face-validity check, not a claim of external validation: the SPS is a decision-support yardstick against which any title, or a developer's own back-catalogue, can be benchmarked, not a measure verified against audited outcomes.

Table 5.2: Top 20 titles by Steam Performance Score. Estimated owners are SteamSpy range midpoints, which explains the repeated round values; review count and sentiment separate titles within a band.

#	Title	Year	Genre	Owners (M)	Reviews	Pos. ratio	SPS
1	Stardew Valley	2016	Indie	35.0	886,195	0.98	97.3
2	RimWorld	2018	Indie	7.5	210,517	0.98	97.2
3	Terraria	2011	Action	35.0	1,409,473	0.97	97.2
4	The Binding of Isaac: Rebirth	2014	Action	7.5	343,420	0.97	97.1
5	Vampire Survivors	2022	Action	7.5	250,334	0.98	97.1
6	Slay the Spire	2019	Indie	7.5	182,277	0.98	97.1
7	Schedule I	2025	Action	15.0	204,041	0.98	97.1
8	Baldur's Gate 3	2023	Adventure	35.0	760,662	0.97	97.1
9	Euro Truck Simulator 2	2012	Indie	15.0	867,404	0.97	97.1
10	Mount & Blade: Warband	2010	Action	7.5	166,405	0.98	97.1
11	Wallpaper Engine	2018	Casual	35.0	894,458	0.98	97.1
12	Satisfactory	2024	Adventure	15.0	232,064	0.97	97.1
13	Factorio	2020	Casual	7.5	215,209	0.97	97.1
14	Hades	2020	Action	7.5	279,741	0.98	97.1
15	Resident Evil 4	2023	Action	7.5	181,708	0.97	97.1
16	Black Myth: Wukong	2024	Action	75.0	1,150,098	0.97	97.1
17	Balatro	2024	Casual	3.5	153,566	0.98	97.0
18	Garry's Mod	2006	Casual	35.0	1,159,707	0.97	97.0
19	Bloons TD 6	2018	Strategy	3.5	353,960	0.97	96.9
20	Cookie Clicker	2021	Casual	3.5	79,892	0.97	96.9

Across the whole catalogue the score is heavily left-skewed (Figure 5.5): the large mass of titles sits at low scores and only a thin upper tail approaches the values in Table 5.2, which is the concentration finding of Chapter 4 re-expressed on a single 0 to 100 scale.

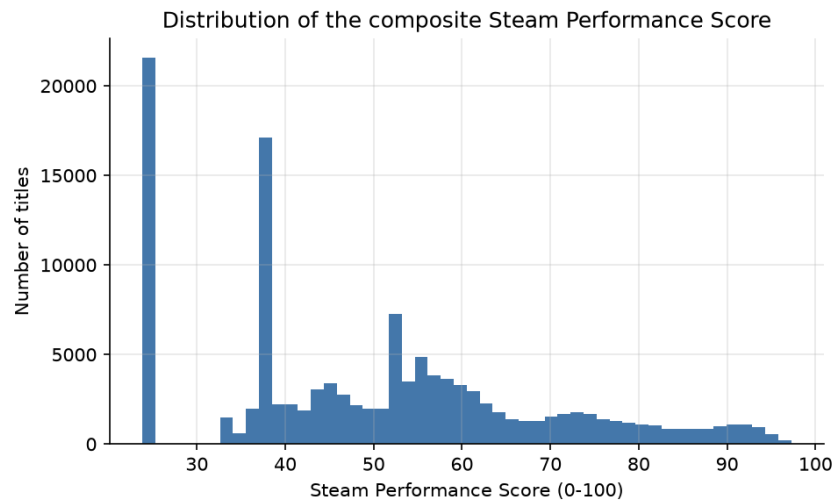


Figure 5.5: Distribution of the Steam Performance Score across all 122,510 titles. The thin upper tail corresponds to the celebrated titles of Table 5.2.

Chapter 6

Discussion

6.1 A superstar market on an open platform

The evidence answers **RQ1** unambiguously: market attention on Steam is a superstar, winner-take-most distribution, not a democratised long tail. Owner and review Gini coefficients of 0.888 and 0.978, and a top-5%-take-80% Pareto structure, sit squarely with Elberse (2008)’s superstar findings and against the optimistic reading of Anderson (2006). This does not refute the long-tail thesis so much as qualify it in the manner of Waldfogel (2017): Steam has an enormously long tail of *supply*, but that tail is commercially very flat, and demand remains concentrated at the head. The still-higher Gini for reviews than for owners is theoretically coherent, since reviewing is a more effortful, self-selecting act than owning, so attention concentrates even more sharply than ownership, echoing Lin et al. (2019).

6.2 Saturation as a structural, not cyclical, phenomenon

The joint trajectory of supply and demand answers **RQ2**: the near-tenfold rise in releases against a seventeenfold fall in owners per release, accompanied by a surge in zero-review titles, is a clear signature of market maturity and saturation (Levitt, 1965; Day, 1981). What makes this particularly significant is that it aligns with Cennamo (2018)’s finding: the decline is more plausibly read as a *structural* consequence of the platform’s openness rather than a temporary cyclical dip. Steam Direct removed the gatekeeping that once limited supply, and Parker, Van Alstyne and Choudary (2016)’s openness–curation trade-off predicts exactly the dilution of per-title attention observed. The Attention Carrying Capacity model (Section 5.2) sharpens this into a measurable claim: the mass-market genres sit at 85 to 91% of their estimated attention ceilings. It is worth being precise about the level at which this operates. The two-way fixed-effects analysis (Section 5.3) and the standardised regression (Section 4.6) agree that the decline

is a *market-wide* effect of time and aggregate supply, not a clean penalty attaching to whichever genre happens to be most crowded in a given year. The robustness checks (Section 4.7) confirm the effect is real rather than a measurement artefact: it survives a review-based ownership proxy and persists among mature cohorts. The counter-movement in publisher HHI, collapsing concentration on the supply side, is the same phenomenon viewed from the other side of the platform: entry has been democratised for producers even as outcomes have concentrated for products. The finding empirically substantiates the practitioner anxieties recorded by the Game Developers Conference (2025) and the “indiepocalypse” discourse analysed by Lipkin (2019).

6.3 Strategic implications of genre crowding

The genre analysis (**RQ3**) localises the saturation. The categories that are most over-supplied relative to demand, Casual, Adventure and Indie, are also those whose release volumes are growing fastest, a self-reinforcing crowding dynamic in which new supply pours into segments least able to absorb it. Genres such as RPG and Strategy remain comparatively under-supplied relative to the demand they command. For a developer, this is directly actionable competitive intelligence: it identifies where the ratio of contestants to available attention is least favourable. The genre caveats of Windleharth et al. (2016) mean these boundaries should be read as broad rather than exact, but the direction of the signal is strong.

To translate this into practical terms: a developer entering the Casual genre in 2025 faces an estimated saturation rate of 85% against the genre’s attention ceiling, and competes with several thousand new releases in that category each year. Under those conditions, the difference between a title that finds an audience and one that does not is rarely a function of production quality alone. Research on review dynamics (Duan, Gu and Whinston, 2008) shows that it is review *volume* rather than review *sentiment* that drives visibility in algorithmic ranking, so the practical priority for a developer in a crowded category is to generate early review activity, whether through Steam Next Fest participation, streamer seeding or launch-period discounting, before the title falls off the new-releases feed. Ma (2025) confirm that Steam-native engagement metrics, including review count, are the strongest predictors of revenue outcomes, outperforming external critic scores. This shifts the strategic conversation from “is the game good?” to “can the game be seen?”, a distinction that the saturation finding makes urgent.

One qualification protects the reader from over-reading it. That a genre is crowded on average does not make it unprofitable in particular. Industry analysis of Steam outcomes stresses that even saturated categories contain titles that earn well, and that execution, timing and marketing separate winners from the mass far more than genre membership does (Carless, 2023; Zukowski, 2024). The crowding index is a statement about base rates, the average contest a new title enters, not a verdict on any individual game. The correct managerial use is as a prior to be updated by a specific title's quality and differentiation, not as a prohibition on entering a popular genre. Read that way, the direction of this study's findings aligns with what the commercial market-intelligence vendors report from independent data: extreme concentration, a hits-driven demand curve, and discoverability rather than production as the binding constraint (Gamalytic, 2024; Video Game Insights, 2024).

It is also worth qualifying the mechanism. The crowding effect described here is consistent with the social-amplification dynamic noted in Section 2: because Steam functions partly as a social and crowdsourcing platform (Thorhauge, 2024; Zanesco, Lajeunesse and French, 2021), a title that fails to cross an early visibility threshold loses algorithmic placement and social momentum simultaneously. This is different from the congestion one might expect in a pure e-commerce setting, where a product can simply sit on a shelf and wait for discovery. On a social platform, the decay in visibility for unremarked titles can be rapid and self-reinforcing, which explains why the share of releases receiving zero reviews has grown so steeply even as the absolute number of active players on Steam has not contracted in proportion.

6.4 Strategic shape of performance segments

The clustering (**RQ4**) gives the distribution a concrete, communicable structure. That a 7.6% Performers segment captures 77.9% of ownership, while a 17.5% Dormant segment captures essentially none, converts the abstract Gini and Pareto statistics into three market realities a manager can reason about. The segmentation aligns with the actionability criteria of Wedel and Kamakura (2000): the groups are identifiable from observable data, substantial in size, and strategically distinct.

6.5 Actionable implications for developers, publishers and platform operators

For developers, visibility rather than production is the binding constraint. With a third of titles receiving no reviews and the median title only seven, the realistic planning assumption is the mid-tail, not the head. Genre selection is itself a concentration decision: entering an over-supplied category such as Casual means competing for a shrinking slice of attention. Since review volume tracks visibility (Duan, Gu and Whinston, 2008) and predicts revenue (Ma, 2025), mechanisms that generate early reviews, such as wishlists, launch discounts and community engagement, are first-order concerns rather than afterthoughts.

For publishers, the value of a publisher lies precisely in moving titles from the mid-tail into the Performers segment through curation, marketing and discoverability, a function that the data show becomes *more*, not less, valuable as the catalogue saturates. A publisher that can reliably identify which of a developer's features are associated with breakout, and deploy marketing resource accordingly, provides an increasingly rare complement to the production capability that many small studios already possess.

For the platform operator, Cennamo (2018)'s diminishing-returns result carries a specific implication that Valve's current revenue model does not fully address. Steam charges developers a fixed publishing fee and takes a percentage share of revenue, a structure that internalises the revenue upside of catalogue growth but not its negative externality: each additional title added to the catalogue marginally dilutes the discoverability of every other title, yet the platform as a whole does not bear that cost. Parker, Van Alstyne and Choudary (2016) frame this as the openness–curation trade-off, and the data here suggest the platform has moved well past the point of optimal openness. Practical responses open to the platform operator include genre-specific release pacing (slowing new entries into the most saturated categories), quality thresholds gated by early-access review rates, and heavier investment in personalised discovery algorithms that match players to relevant titles from the long tail of supply rather than surfacing only the already-prominent head. These interventions would move the platform in the direction that the prescriptive analytics tradition (Lepenioti et al., 2020) targets: from describing and diagnosing what has happened to actively shaping what happens next.

Chapter 7

Conclusion

7.1 Summary of findings

This dissertation set out to characterise, through a Business Intelligence lens, how performance is distributed across Steam-distributed video games and whether the marketplace shows signs of saturation. Across 122,510 titles it resolves each of its five research questions in turn.

The picture that emerges is consistent across all five research questions. Market attention on Steam is a superstar, winner-take-most distribution (**RQ1**): the owner Gini coefficient sits at 0.888, and the top 5% of titles hold 81% of all ownership. The platform is saturating structurally rather than cyclically (**RQ2**), with a tenfold rise in annual releases accompanied by a seventeenfold fall in owners per release. On **RQ3**, over-supply is concentrated in the Casual and Adventure genres, which are also growing fastest, while RPG and Strategy remain comparatively under-supplied. On **RQ4**, the catalogue resolves into a small dominant Performers segment (7.6% of titles, 77.9% of ownership), a large Modest mid-tail (75.0% of titles, 22.1% of ownership), and a substantial Dormant segment (17.5% of titles, negligible ownership). And on **RQ5**, a diagnostic regression finds that release timing, not any controllable title attribute, is most strongly associated with reach, which brings the analysis back to saturation as the organising fact of the marketplace.

7.2 Academic and practical contributions

Empirically, the study delivers a current, whole-catalogue quantification of concentration and saturation on the dominant PC games platform, hardened by robustness checks that reproduce the headline results under an independent ownership proxy and among mature cohorts. Methodologically, it introduces the Attention Carrying Capacity framework, a novel adaptation of the Michaelis–Menten/Monod saturation law to genre-level demand, fitting at a median R^2

of 0.99, together with a diagnostic regression, the Genre Saturation–Opportunity Matrix and a breakout classifier that holds a ROC-AUC of 0.89 on a forward-in-time hold-out and is made interpretable with SHAP. Together these show how concentration indices, Pareto analysis, saturation modelling and machine learning can be integrated into a reproducible BI decision-support framework transferable to other digital platform marketplaces such as app stores and streaming catalogues. The two-way fixed-effects specification further establishes that saturation operates at the market level rather than the genre level. Practically, the study converts these measures into specific guidance for developers, publishers and the platform.

Beyond its specific findings, the framework developed here has a broader transferability. The combination of concentration measurement, a Michaelis–Menten attention ceiling, and a validated breakout classifier is not inherently Steam-specific; it applies wherever a digital platform lists a large number of competing items and engagement data are publicly observable. Mobile app stores, streaming music catalogues and e-book platforms all exhibit the same structural tension between open supply and concentrated demand, and the same instruments could in principle be applied to characterise saturation and guide positioning decisions in those markets.

The temporal validation of the breakout classifier also merits emphasis as a methodological contribution in its own right. Evaluating a model trained through 2022 on titles released from 2023 onward, with no leakage between the two windows, is the only evaluation design that answers the question a practitioner actually cares about: will the patterns the model has learnt still hold for games that have not shipped yet? That the ROC-AUC is essentially unchanged between the cross-validated estimate (0.887) and the forward-in-time hold-out (0.886) offers meaningful evidence of out-of-sample robustness, going beyond the within-sample fit that most published classification studies report.

Practically, the Genre Saturation–Opportunity Matrix and the Steam Performance Score can be used by a developer or publisher on an ongoing basis, not merely as a one-time diagnostic. The Matrix is designed to be recalculated as new release cohorts accumulate, so that a studio monitoring the Simulation quadrant, which is currently the lowest-saturation major genre, can track whether it fills before its own title ships. The SPS enables a developer to benchmark any title in its back-catalogue against the full 122,510-game distribution, identifying which dimensions of performance are lagging and where the most actionable improvement lies.

7.3 Methodological limitations and future research

The limitations of Section 3.8 bound the claims: ownership figures are modelled estimates whose SteamSpy provenance degraded after 2018, the most recent cohorts are right-censored, and genre labels are a folksonomy. A further limitation is one of identification: because same-genre releases rose in lockstep with the whole platform, the crowding elasticity cannot be separated cleanly from the market-wide time trend, so this study attributes the per-title decline to platform-level saturation rather than to genre-specific congestion. Resolving that entanglement is the clearest line for future work. A panel that follows individual titles over time, or an instrument for genre-level supply that is independent of the aggregate trend, could recover a causal crowding effect that this cross-section cannot. Beyond identification, future work could triangulate the estimated-owners proxy against disclosed revenues where available, incorporate wishlist and follower data to model the discovery funnel directly, extend the release-cohort analysis into an explicit diffusion (Bass) model of platform maturity (Bass, 1969), and complement the cross-sectional clustering with a longitudinal study of how titles migrate between segments over their life cycle.

A broader limitation concerns the relationship between the patterns this study documents and the causal mechanisms that produce them. The findings are primarily descriptive and diagnostic: they characterise the structure of the market at a point in time and trace how that structure has evolved, but they do not establish why any given developer's title attracted the owners it did, nor do they rule out confounding explanations for the aggregate decline. The regression of Section 4.6 is explicit about reading its coefficients as associations rather than effects, and the two-way fixed-effects analysis of Section 5.3 demonstrates the difficulty of isolating a causal crowding penalty from the market-wide time trend. This is an acknowledged limitation of cross-sectional, observational designs. Stronger theoretical insight into the underlying mechanisms, for instance whether the decline in per-title attention is driven by competition for algorithmic placement, by shifts in player behaviour, or by changes in Valve's curation policies, would require either an experimental design (which the archival nature of the data precludes) or a structural model that makes explicit assumptions about how developers and players respond to market conditions. Developing such a model, informed by the empirical patterns documented here, is a natural and important direction for future research. In particular, a structural model of developer entry and exit decisions, combined with a model of player search and discovery, could

shed light on whether the saturation dynamic is self-correcting (as developers exit unprofitable categories) or self-reinforcing, as the evidence of continued rapid growth in over-supplied genres suggests. The present study provides the empirical characterisation that such theoretical development would need to discipline and validate.

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Appendix A

Data Dictionary (Retained Variables)

Variable	Type	Description
app_id	integer	Unique Steam application identifier
name	string	Title of the game
release_date	date	Store release date
release_year	integer	Calendar year of release (analysis cohort)
owners_low / owners_high	integer	Bounds of the estimated-owners range
owners_mid	numeric	Midpoint of the estimated-owners range (reach proxy)
peak_ccu	integer	All-time peak concurrent users
price	numeric	Current store price (USD)
positive / negative	integer	Count of positive / negative user reviews
total_reviews	integer	positive + negative
review_ratio	numeric	positive ÷ total_reviews (sentiment)
is_free	boolean	Price equals zero
revenue_proxy	numeric	owners_mid × price (Boxleiter-style proxy)
avg / median_playtime_forever	integer	Mean / median lifetime playtime (minutes)
primary_genre	string	First listed genre
developers / publishers	string	Studio / publisher names
genres / categories / tags	string	Descriptive metadata
cluster	integer	Assigned K-Means performance segment

Appendix B

Technical Reproducibility and Computational Pipeline

The full analysis regenerates deterministically from the raw dataset via an ordered sequence of five scripts, each writing its outputs to a results directory and a figures directory before the next script is run. All randomised procedures, including the K-Means initialisation and the bootstrap resampling, are seeded with a fixed value (seed = 42) so that every statistic, table entry and figure in this dissertation can be regenerated exactly. The predictive components apply standard leakage controls: the breakout classifier is trained only on features observable at or before release, its cross-validation folds are stratified by the outcome to preserve the base rate, and its temporal hold-out strictly separates titles released up to 2022 from the 2023-onward test set so that no post-release information or future cohort contaminates the training data.

1. `analysis/01_clean_data.py` – ingestion, cleaning and derivation of variables; writes the cleaned data/games_clean.parquet file.
2. `analysis/02_bi_analysis.py` – descriptive statistics; concentration analysis (Gini, Lorenz, HHI); Pareto analysis; trend analysis; genre saturation; K-Means clustering.
3. `analysis/03_advanced_analytics.py` – bootstrap concentration intervals; the Attention Carrying Capacity model; the crowding elasticity; the Genre Saturation–Opportunity Matrix; the breakout classifier (with temporal validation and SHAP); the Steam Performance Score.
4. `analysis/04_regression.py` – the multivariate OLS diagnostic with robust standard errors and VIF diagnostics, together with the paid-only and standardised-coefficient robustness runs.
5. `analysis/05_robustness.py` – ownership triangulation via the Boxleiter proxy; the two-way fixed-effects elasticity; the maturity check; the Gini-over-time series.

Every headline figure reported in this dissertation is collated in a single `results/numbers_ledger.md` file, so each statistic can be traced back to the script and output that produced it. The computational envi-

ronment requires Python 3 with pandas, NumPy, scikit-learn, SciPy, statsmodels, SHAP and Matplotlib.

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